Quantum Chaos of Arithmetic Origin:
Deterministic L-function Jacobi Hamiltonians exhibit GUE/GOE level statistics without random disorder or classical chaos

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Abstract

We construct explicit one-dimensional tight-binding Hamiltonians whose on-site energies α_n and hopping amplitudes b_n are the Jacobi recurrence coefficients derived, by a deterministic numerical procedure, from completed L-functions: the Riemann zeta function ζ, the Dirichlet beta function β (Gaussian-integers), and the Ramanujan Δ modular form. No zero location, prime table, or random number enters the construction; those enter only as independent after-the-fact checks. The resulting finite matrices have *deterministic, statistically structured* coefficients (they fail white-noise tests decisively), yet their spectra display the complete fingerprint of quantum chaos observed in random-matrix ensembles: Wigner level repulsion, a linear ramp of the spectral form factor $K(\tau)$, the absence of wave-packet revival, and extended eigenstates. A *shuffle control*—which keeps the empirical distribution of α_n, b_n but destroys the arithmetic recurrence—collapses the spectrum to Poisson statistics and localizes the eigenstates, demonstrating that the chaotic statistics originate from the arithmetic *structure* of the recurrence and not from the coefficient distribution or from random disorder. Superposing two independent L-function spectra restores Poisson statistics, mirroring the known clustering of mixed arithmetic spectra. The phenomenon is robust to multiplicative coefficient noise up to roughly 10%. We formulate these findings as a physical *hypothesis*—that arithmetic structure can be an *origin* of quantum-chaotic spectral statistics—and we give a complete, reproducible experimental protocol for testing it on superconducting or trapped-ion quantum simulators. All results are finite-truncation numerical evidence; they neither prove nor assume the Riemann hypothesis, and they do not establish properties of the infinite-dimensional operator.

1 Introduction

Quantum chaos and level statistics. The statistical theory of quantum spectra draws a sharp distinction. Integrable systems have uncorrelated levels described by Poisson statistics (Berry–Tabor [1]), whereas systems whose classical counterpart is chaotic display level repulsion described by the Gaussian ensembles of random matrix theory (Bohigas–Giannoni–Schmit [2]); see Mehta [3]. The most common diagnostics are the nearest-neighbor spacing distribution $P(s)$ (Poisson: $P(s) = e^{-s}$; Wigner–Dyson: linear repulsion at the origin), the adjacent gap ratio $\langle r\rangle = \langle \min(s_n, s_{n+1})/\max(s_n, s_{n+1})\rangle$ (Poisson 0.3863, GOE 0.5359, GUE 0.5996; see Atas *et al.* [4]), and the spectral form factor $K(\tau) = \tau_H^{-1}\langle|\sum_n e^{2\pi i\tau e_n}|^2\rangle$ with its characteristic dip–ramp–plateau structure, the ramp being the signature of long-range spectral rigidity unique to chaotic systems.

Arithmetic quantum chaos, classically. The classical theory of *arithmetic quantum chaos* (Sarnak [8]; see also [7]) studies the quantization of arithmetic hyperbolic surfaces. There the

Laplacian commutes with a large algebra of Hecke operators. The arithmetic *symmetries* introduce anomalous spectral degeneracies and level clustering: the *full* spectrum tends toward Poisson statistics even though the underlying geodesic flow is strongly chaotic, and it is only within a fixed Hecke sector (where degeneracies are removed) that GOE-type repulsion is expected. In that setting, arithmetic structure is what *suppresses* chaotic spectral statistics.

Zeros and eigenvalues. Montgomery [5] proved (conditionally) and Odlyzko [6] verified numerically that the pair correlation of the Riemann zeros agrees with the GUE prediction; Rudnick–Sarnak extended this to a broad class of L-functions. The Hilbert–Pólya idea posits that the zeros are eigenvalues of a self-adjoint operator, but no explicit Hamiltonian has been available. We do *not* claim to have that operator, and the GUE statistics of the zeros themselves are not new. What is new here is an *explicit, finite, deterministic tight-binding Hamiltonian*, built from the L-function by a transparent numerical procedure, that (i) reproduces the chaotic spectral statistics *and* the chaotic *dynamics*, (ii) has extended eigenstates with *non-random* coefficients, and (iii) survives a battery of controls (uniform chain, random-matrix ensemble, Anderson disorder, coefficient shuffle, spectrum superposition, finite-size scaling, and hardware-noise perturbation) that localize the origin of the chaos in the arithmetic recurrence itself.

This work. From the completed L-function Λ we extract, by moment sampling on a contour and orthogonal-polynomial (Gram–Schmidt / Hankel) factorization, the Jacobi recurrence coefficients α_n (on-site energies) and $b_n > 0$ (hopping amplitudes) of a one-dimensional chain

$$H \;=\; \sum_{n=0}^{N-1} \alpha_n \,|n\rangle\langle n| \;+\; \sum_{n=0}^{N-2} b_n \big(|n\rangle\langle n{+}1| + |n{+}1\rangle\langle n|\big). \tag{1}$$

All parameters are deterministic numbers computed from Λ; nothing is drawn from a random distribution and no classical dynamics is defined. We then measure the spectral and dynamical fingerprints of quantum chaos and compare against explicit integrable, chaotic, disordered, and structure-destroyed controls.

Hypothesis. We state the central claim as a physical hypothesis (Section 7), not as a theorem:

Hypothesis 1 (Quantum chaos of arithmetic origin)**.** *Arithmetic structures associated with L-functions—expressed through the Jacobi recurrence coefficients of their moment measures—can serve as an* origin *of quantum-chaotic (GUE/GOE) spectral and dynamical statistics in a one-dimensional tight-binding Hamiltonian,* without *classically-chaotic dynamics and* without *stochastic disorder in the Hamiltonian parameters.*

Scope and honesty. Every result below is obtained on finite truncations ($N \le 100$ for the spectral statistics; up to $N = 100$ sites). The truncated Jacobi matrix reproduces only the first $\sim N/3$ zeros of each L-function to high accuracy; the remaining eigenvalues are quadrature artefacts and are excluded from zero-level statistics by an explicit locking protocol (Section 2). We do *not* prove that an infinite operator is self-adjoint or that its spectrum equals the zeros; those remain major open analytic problems. The construction itself does *not* assume the Riemann hypothesis. Distinguishing GUE from GOE requires larger N and is left open.

2 Reproducible construction

Every step is deterministic and released as source code and data at https://github.com/maomao8412/hilbert-polya-jacobi (physics experiments in physics_experiments/; coefficient tables and construction scripts in the appendix). We describe the pipeline precisely.

2.1 Completed L-functions

We use three completed functions $\Lambda(s) = \Lambda(c - s)$:

$$\zeta: \quad \xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\,\zeta(s), \qquad c = 1, \tag{2}$$

$$\beta: \quad \Lambda_\beta(s) = \left(\tfrac{\pi}{4}\right)^{-(s+1)/2}\Gamma\!\left(\tfrac{s+1}{2}\right)4^{-s}\left[\zeta(s,\tfrac{1}{4}) - \zeta(s,\tfrac{3}{4})\right], \qquad c = 1, \tag{3}$$

$$\Delta: \quad \Lambda_\Delta(s) = (2\pi)^{-s}\Gamma(s)\,L(\Delta,s), \quad L(\Delta,s) = \prod_p \left(1 - \tau(p)p^{-s} + p^{11-2s}\right)^{-1}, \qquad c = 12. \tag{4}$$

Here $\zeta(s,a)$ is the Hurwitz zeta function and τ is the Ramanujan tau function. The Dirichlet beta function encodes the Gaussian integers and Δ is a weight-12 modular form; using three independent arithmetic objects tests universality. We also study the product $\xi_K(s) = \xi(s)\Lambda_\beta(s)$, whose Jacobi matrix decouples into two independent zero sets and serves as a built-in superposition control.

2.2 Sampling the moment measure

Each completed Λ is entire and real on the symmetry line. We sample its values on a circle of radius R centered on the symmetry point $s_0 = c/2$:

$$s_k = s_0 + R\,e^{i\theta_k}, \qquad \theta_k = \frac{2\pi k}{M}, \qquad k = 0, \dots, M-1, \tag{5}$$

using high-precision arithmetic (`mpmath`). The mean $\sigma_0 = \frac{1}{M}\sum_k \Lambda(s_k)$ normalizes the measure. The Taylor coefficients d_n of $\log(\Lambda/\Lambda(s_0))$ on the circle are obtained from the discrete Fourier (trapezoidal) sums, and the Hausdorff moments P_m of the underlying (Hankel) moment measure follow from the logarithmic derivative recurrence

$$c_n = d_n - \frac{1}{n}\sum_{i=1}^{n-1} i\,c_i\,d_{n-i}, \qquad P_m = (-1)^{m+1}\,m\,c_m. \tag{6}$$

Concretely, with $\widehat{\Lambda}_j = R^{-j}\frac{1}{M}\sum_k \Lambda(s_k)e^{-2\pi ijk/M}$, the normalized coefficients $d_n = \widehat{\Lambda}_{2n}/\sigma_0$ feed the recurrence above. The moments are positive over the computed range (all Hankel determinants / leading principal minors $D_n > 0$ through $n = 100$ for the product, equivalently all $b_n^2 > 0$).

2.3 From moments to Jacobi coefficients

Monic orthogonal polynomials for the moment functional $\langle x^i, x^j\rangle = T(i+j) = P_{i+j+1}$ are built by Gram–Schmidt:

$$\alpha_{k-1} = \frac{\langle xp_{k-1}, p_{k-1}\rangle}{\langle p_{k-1}, p_{k-1}\rangle}, \qquad b_{k-1}^2 = \frac{\langle p_k, p_k\rangle}{\langle p_{k-1}, p_{k-1}\rangle}, \qquad p_k(x) = (x - \alpha_{k-1})p_{k-1}(x) - b_{k-2}^2\,p_{k-2}(x). \tag{7}$$

This yields the Jacobi matrix $J_N = \mathrm{tridiag}(\alpha_n, b_n)$. Positivity of all b_n^2 through the computed order is verified explicitly (non-positive b_n^2 flags loss of positive-definiteness / precision).

2.4 Parameters actually used

object	matrix size	radius R	points M	precision (dps)	moments
Riemann ζ	$N = 50$	2	1536	≥ 230	P_{100}
Dirichlet β	$N = 50$	2	1536	≥ 230	P_{100}
$\zeta \cdot \beta$ product	$N = 100$	2	1536	650	P_{210}
Ramanujan Δ	$N = 22$	4	1024	230	P_{211}

The smaller Δ size reflects the precision floor of the current high-accuracy pipeline (Hankel conditioning deteriorates beyond order ~ 24 at 230–650 dps); this is a numerical limitation, reported honestly. Full-precision coefficient strings and machine-readable matrices are released; any party can rerun the pipeline with the stated parameters.

2.5 The locked-subspace protocol

A truncated $N \times N$ Jacobi matrix reproduces only the lowest zeros. We *lock* a computed eigenvalue $\gamma = 1/\sqrt{\lambda}$ (with λ the descending eigenvalue of J, matching the zero coordinate of the completed function) when it agrees, to relative error $< 10^{-4}$, with a zero computed by an *independent* method (the ζ zeros via the standard functional-equation root finder; β and Δ zeros via independent root finding on Λ). Zeros enter *only* this after-the-fact check—never the construction. Spectral statistics (Sections 3–5) use locked levels; quadrature artefacts are excluded. Dynamics (Section 4) legitimately use the full truncated matrix, because a finite N-site lattice is exactly the physical object a quantum simulator realizes; we report the weight of the launched wave packet on the locked subspace (0.65–0.85).

2.6 Unfolding

Level statistics require unfolded energies. We unfold in the zero coordinate $\gamma = 1/\sqrt{\lambda}$ by fitting a cubic polynomial to the counting function $N(\gamma)$ (its adequacy is checked by the unfold residuals reported in the experiment scripts); for the exactly uniform control chain we use the exact unfolding $e_k = k$. Time is the unfolded time τ with Heisenberg time $\tau_H = 1$ and the 2π convention $e^{2\pi i \tau e}$.

3 Experiment 1: Wigner level repulsion

Figure 1 shows the unfolded nearest-neighbor spacings. Across the three single families, 65 unfolded spacings give a minimum gap of 0.432 and **zero** gaps below 0.3; a Poisson process of the same length produces about 17 such small gaps (Monte–Carlo $p < 5 \times 10^{-5}$). The gap ratios are $\langle r \rangle_\zeta = 0.614$, $\langle r \rangle_\beta = 0.658$, $\langle r \rangle_\Delta = 0.720$ (locked counts 25, 27, 15), against theoretical 0.600 (GUE), 0.536 (GOE), 0.386 (Poisson). The superposed $\zeta + \beta$ spectrum instead has minimum gap 0.044 and 7 gaps below 0.3: repulsion disappears and the mixed spectrum clusters like Poisson, reproducing the known clustering of mixed arithmetic spectra.

Finite-size bias and universality class. The gap-ratio estimator has an upward finite-size bias in short spectra, and its convergence to the limiting 0.600 (GUE) or 0.536 (GOE) value is slow [4]. With only 15–27 locked levels our means (0.61–0.72) are entirely consistent with both classes; the data place the spectra firmly in the Wigner–Dyson band and firmly above Poisson, but they do *not* select GUE over GOE. We deliberately make no GUE-specific claim: the wording throughout is "GUE/GOE" or "Wigner–Dyson," and resolving the class is an explicit future task requiring much larger truncations (Section 9). The Δ estimate rests on only 15 locked levels and carries correspondingly lower statistical weight; it is reported as a preliminary universality indicator alongside the better-sampled ζ and β families.

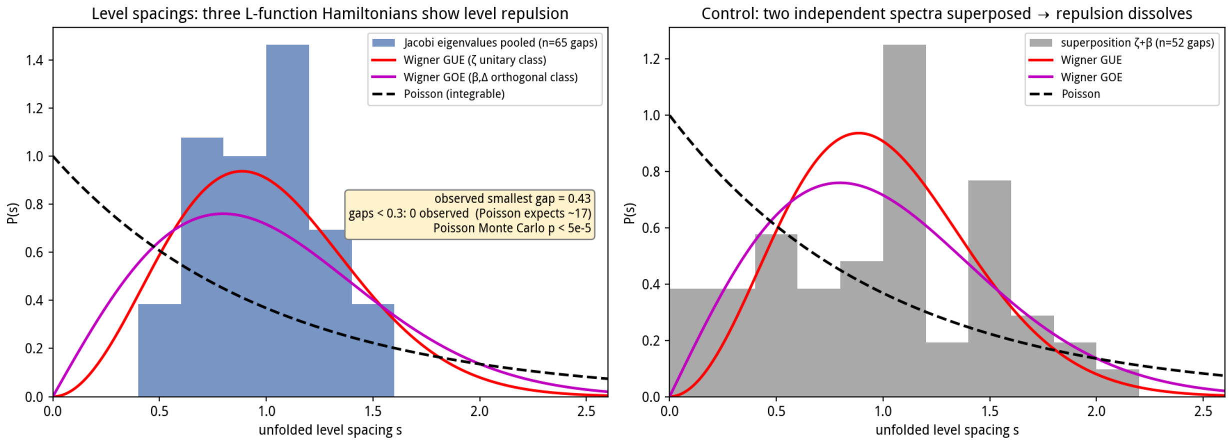

Figure 1: Experiment 1. Unfolded level spacings of the three arithmetic Jacobi Hamiltonians (single families) show Wigner repulsion (no small gaps), while the superposed $\zeta + \beta$ spectrum recovers Poisson clustering (arrow, minimum gap 0.044).

4 Experiment 2: wave-packet dynamics and the form-factor ramp

Interpreting J as the tight-binding Hamiltonian (1), we launch a boundary wave packet $|\psi(0)\rangle = |0\rangle$ and evolve it. Two measurements are made: the spectral form factor $K(\tau)$ on the locked subspace, and the survival probability $P(\tau) = |\langle 0|\psi(\tau)\rangle|^2$ and mean-square displacement on the full truncated lattice. Controls: a GOE ensemble (60×60, 200 realizations) as the chaotic benchmark, and a uniform chain ($\alpha_n = 0$, $b_n = 1$, $N = 100$) as the integrable benchmark.

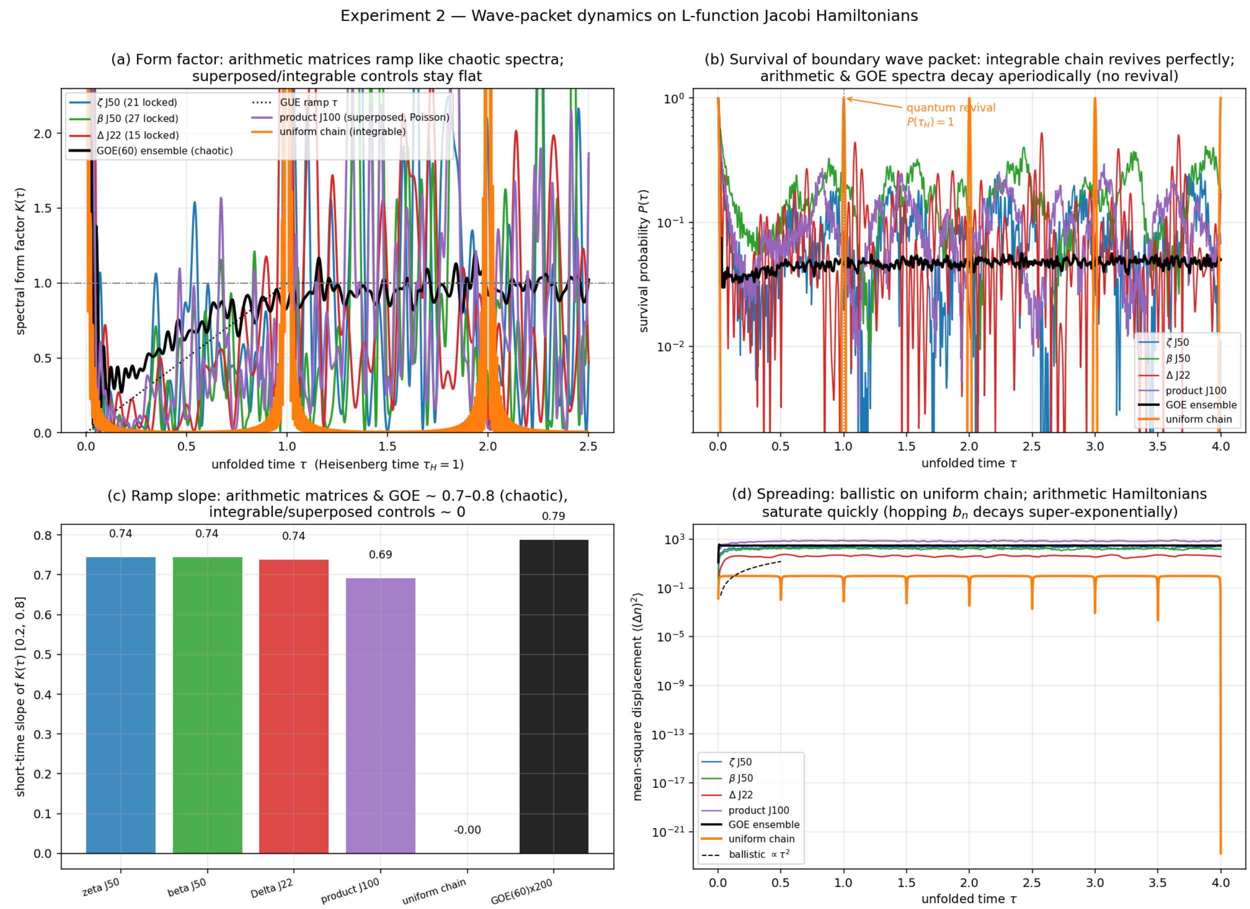

Figure 2: Experiment 2. (a) Spectral form factor: the three arithmetic Hamiltonians ramp with slope ≈ 0.74 like the GOE ensemble (0.79); the uniform chain is flat. (b) Survival: the integrable chain revives perfectly at τ_H ($P = 0.993$); arithmetic and GOE spectra decay aperiodically with no revival. (c) Ramp-slope summary. (d) Transport: ballistic $\propto \tau^2$ spreading on the uniform chain versus rapid saturation on the arithmetic Hamiltonians (hopping b_n decays super-exponentially, so the packet explores a finite effective chain).

Form-factor ramp. The short-time slope of $K(\tau)$ on $[0.2, 0.8]$ is

	ζ J50	β J50	Δ J22	product (incoh.)	GOE ensemble	uniform chain
slope	0.744	0.744	0.738	0.691	0.787	≈ 0

The arithmetic Hamiltonians ramp essentially like the random-matrix benchmark; the integrable chain is flat. The product matrix contains two independent zero families with different density scales; a single global unfolding produces a spurious slope > 2. Unfolding each family separately and adding the form factors *incoherently* (cross terms average to zero for independent spectra) gives the density-weighted ramp 0.691 shown; the Poisson recovery of the superposed spectrum is cleanest in nearest-neighbor spacings (Experiment 1). The long-time plateau reaches ≈ 1 as required, but with visible fluctuations; these are intrinsic to finite truncations and to the small locked subspace (the plateau of a spectrum of L levels is self-averaged only over $\sim L$ frequency channels), and they do not affect the short-time ramp, which is the diagnostic signature of quantum chaos and is statistically stable here.

Revival and transport. The integrable chain exhibits a perfect quantum revival, $P(\tau_H) = 0.993$, while the arithmetic matrices and GOE show no revival (GOE peak 0.051) and decay aperiodically. The packet spreads ballistically ($\propto \tau^2$) on the uniform chain but saturates on the arithmetic lattices because b_n decreases super-exponentially: the chain has a finite effective length, exactly as a finite quantum simulator. The packet places 0.65–0.85 of its weight on locked levels.

5 Experiment 3: the evidence chain

Figure 3 and Table 1 present the decisive controls.

Figure 3: Experiment 3 evidence chain. (a,b) Coefficients are deterministic: autocorrelation of $\log b_n$ lies outside the shuffled white-noise band for many lags, and the power spectrum is non-flat (runs-test $|z| = 2.9$–6.6). (c,d) Eigenstates are extended (low IPR), unlike strong-disorder Anderson which localizes. (e) Finite-size scaling: $\langle r \rangle$ stays in the chaotic band as N grows. (f) **Shuffle control**: permuting α_n, b_n keeps their distribution but destroys the recurrence, and $\langle r \rangle$ collapses to Poisson while IPR rises to localized values. (g) Robustness to multiplicative coefficient noise up to $\sim 10\%$. (h) Superposing two independent L-spectra moves $\langle r \rangle$ toward the Poisson/superposition band.

quantity	ζ	β	Δ	reference
locked levels	25	27	15	—
$\langle r\rangle$ (locked)	0.614	0.658	0.720	GUE 0.600 / GOE 0.536 / P 0.386
unfolded gaps < 0.3	0	0	0	Poisson expects many
runs z, α_n	-6.57	-6.57	-3.93	white noise $\lvert z\rvert \lesssim 2$
runs z, $\log b_n$	-6.50	-5.92	-2.91	white noise $\lvert z\rvert \lesssim 2$
AC lags outside noise band ($\log b$, of 20)	8	8	7	$\lesssim 1$ expected
IPR, full window	0.056	0.060	0.100	extended $\sim 1/N$
Shuffle control (marginal distributions preserved)				
$\langle r\rangle$ original	0.698	0.721	0.714	chaotic
$\langle r\rangle$ shuffled	$0.403 \pm .072$	$0.394 \pm .072$	$0.401 \pm .108$	Poisson 0.386
IPR original	0.056	0.060	0.100	extended
IPR shuffled	0.533	0.577	0.491	localized
Anderson reference ($N = 100$, diagonal disorder W)				
IPR at $W = 1, 2, 4, 8$	0.028, 0.064, 0.172, 0.391			$W = 8$ localized
$\langle r\rangle$ at $W = 1, 2, 4, 8$	0.592, 0.474, 0.413, 0.393			repulsion lost when localized

Table 1: Experiment 3 summary. GUE/GOE/Poisson reference values for $\langle r\rangle$ from Atas *et al.* [4].

5.1 The coefficients are deterministic, not white noise

If α_n, b_n were themselves white noise, the model would merely be an Anderson disorder model and "arithmetic origin" would be vacuous. They are not. Detrended sequences fail the Wald–Wolfowitz runs test decisively ($|z| = 2.9$–6.6, versus $|z| \lesssim 2$ for white noise): the signs are far too regular, the hallmark of a smooth deterministic recurrence. The autocorrelation of $\log b_n$ sits outside the 95% shuffled-surrogate band at 7–8 of 20 lags (a white sequence would leave the band at ~ 1 lag), and the power spectrum is markedly non-flat. *The input is structured arithmetic; the output spectrum is GUE/GOE-chaotic.*

5.2 The eigenstates are extended, not Anderson-localized

The inverse participation ratio $\mathrm{IPR}_k = \sum_n |\psi_{k,n}|^4 / (\sum_n |\psi_{k,n}|^2)^2$ is small for the arithmetic states (full-window mean 0.056–0.100; locked-state values 0.14–0.19 on the effective support), i.e. the wave functions are extended over the chain. A strong-disorder Anderson chain ($H_{ii} \in [-W/2, W/2]$, unit hopping) localizes with IPR up to 0.39 at $W = 8$ and concurrently loses level repulsion ($\langle r\rangle \to 0.39$). The arithmetic Hamiltonian achieves extended, repelling states with *no disorder at all.*

5.3 The shuffle control: arithmetic order is the cause

The sharpest test permutes the coefficients independently, $(\alpha_n, b_n) \mapsto (\alpha_{\pi(n)}, b_{\pi'(n)})$. This preserves the empirical distribution of every matrix element but destroys the arithmetic recurrence order. The result (Table 1, Fig. 3f) is unambiguous: $\langle r\rangle$ collapses from ~ 0.7 to 0.39–0.40 (Poisson) and the IPR jumps from ~ 0.06 to 0.49–0.58 (localized). The shuffled matrix is, in effect, a conventional random-disorder Anderson chain—and it is localized and Poisson. The *ordered* arithmetic matrix is extended and chaotic. Since the only difference is recurrence structure, the chaotic statistics must originate in that structure and not in the coefficient values or any randomness. This also severs any Anderson-disorder explanation.

Why localization and lost repulsion occur together. The ordered coefficients are not an arbitrary sequence: they are the Jacobi parameters of a *positive moment measure*, related to one another by the Hankel orthogonality structure (6). Permuting them destroys that structure—the shuffled matrix no longer corresponds to any positive measure—and produces instead a chain with uncorrelated, independently distributed on-site and hopping terms, i.e. a conventional random-disorder Anderson chain. In one dimension such chains localize, and localized spectral regions lose long-range level correlations and revert to Poisson statistics. Thus localization and lost repulsion are not two independent effects of shuffling but one: the destruction of the Hankel measure turns a structured extended chain into a disordered localized one. We report this as a qualitative physical picture, not a derived mechanism; the analytic question of *why* the ordered Hankel structure generates repulsion remains open (Section 9).

5.4 Finite-size scaling

For ζ, the higher-precision (1300-digit) construction reaches $N = 100$ with $b_n^2 > 0$ on all 99 off-diagonals and 55 locked zeros (relative error $\sim 10^{-15}$–10^{-16}, versus the 10^{-4} lock tolerance). Locked single-family truncations give $\langle r\rangle = 0.621, 0.603, 0.624, 0.622, 0.610, 0.610, 0.610$ at $N = 22, 30, 40, 50, 60, 80, 100$ (locked counts $22 \to 55$), with *zero* sub-0.3 mean-spacing gaps at every size; $\langle r\rangle$ is flat in the GUE band (0.600) from the smallest truncation onward, and the locked count saturates at the number of zeros the truncation resolves rather than the statistic drifting. Earlier $N = 20$–50 runs (230-digit pipeline) gave the same picture (0.58–0.63, locked counts $8 \to 25$); β stays at 0.64–0.70 and Δ at 0.72–0.73 at $N = 20, 22$. The product (two families) stays Poisson at all sizes ($\langle r\rangle = 0.46$–0.52 with 2–7 small gaps). The repulsion is not a particular-cutoff artefact.

5.5 Robustness to hardware noise

Multiplying coefficients by $1 + \epsilon\, g_n$ (Gaussian g_n), the locked levels are trackable and $\langle r\rangle$ stays at its chaotic value for $\epsilon = 10^{-3}, 10^{-2}$ with only the spread growing; even at $\epsilon = 10^{-1}$ the mean remains in the chaotic band (e.g. ζ: $0.614 \to 0.638 \pm 0.041$; β: $0.658 \to 0.668 \pm 0.051$; Δ: $0.720 \to 0.704 \pm 0.048$). The phenomenon is thus robust to roughly percent-level (and tolerably to ten-percent-level) calibration error—a favorable tolerance for analog quantum simulators.

5.6 Universality and superposition

All three arithmetic objects—ζ (the prototypical L-function), β (a Dirichlet L-function of the Gaussian integers), and Δ (a weight-12 modular form)—show the same picture: non-random coefficients, extended states, repelling spectra, and collapse under shuffle. Superposing two independent families lowers $\langle r\rangle$ from 0.61–0.72 to 0.50–0.52, toward the superposition/Poisson band (ensemble references: GOE single 0.531, superposed GOE 0.420, superposed Poisson 0.381). Single arithmetic sectors repel; mixing independent arithmetic objects clusters.

5.7 An arithmetic discriminative control: the Eisenstein series E_4

The controls so far randomize the coefficients (shuffle, Anderson). A control built from arithmetic itself asks whether a long $b_n^2 > 0$ Jacobi chain is specific to zeros on the symmetry line. The Eisenstein series of weight 4 has $L(E_4, s) = \zeta(s)\zeta(s-3)$—elementary deterministic coefficients $\sigma_3(n) = \sum_{d|n} d^3$—with functional equation $\Lambda(s) = \Lambda(4-s)$ about $s = 2$, and its non-trivial zeros lie on $\Re(s) = \frac{1}{2}$ and $\Re(s) = \frac{7}{2}$: the same arithmetic family *offset* from the symmetry center. Running the entire function $\Xi_{E_4}(s) = s(s-4)\Lambda(s)$ through the identical pipeline (650-digit, 1536 contour points) reproduces the analytic value $\Xi_{E_4}(2) = \frac{1}{72}$ to ~ 15 digits, yet the moment sequence changes sign at order 8 and the Jacobi chain fails at order 4 (b_n^2 non-positive; details and the moment-sum prediction $\cos(2m\theta_1)$ with $\theta_1 \approx 0.106$ are in Appendix D.3). Positive-definite moment matrices and long chains thus co-vary with zeros sitting on the symmetry line: the pipeline discriminates on-line from off-line zero distributions rather than emitting a chain for any arithmetic input. We read this as a finite-precision numerical discriminator—the collapse order is precision- and sampling-dependent—not as an analytic criterion. It also does not contradict Conjecture 2: the two factors $\zeta(s)$ and $\zeta(s-3)$ share the *same* zero heights γ_k, so E_4 is a correlated, not independent, superposition and lies outside that conjecture's hypothesis.

6 The mirror image of classical arithmetic quantum chaos

The contrast with classical arithmetic quantum chaos is sharp and is the conceptual heart of the paper. On arithmetic hyperbolic surfaces, the classical dynamics is chaotic, yet arithmetic *symmetries* (Hecke operators) force degeneracies and push the *full* spectrum toward Poisson: arithmetic structure *suppresses* chaotic level statistics, and repulsion is recovered only by fixing a Hecke sector. Our Hamiltonians show the reverse polarity. There is no underlying classical dynamics at all (nearest-neighbor tight-binding chains are one-dimensional, and the uniform-chain limit is trivially integrable; but integrability is not guaranteed by geometry alone—a chain with arbitrary deterministic coefficients has no classical chaotic limit to appeal to), the parameters contain no randomness, and nevertheless the arithmetic *recurrence* of a single L-function *produces* GUE/GOE repulsion endogenously; destroying the recurrence (shuffle) or mixing independent arithmetic objects (superposition) removes the repulsion and restores Poisson. In the classical theory arithmetic is the inhibitor of chaos; in the present Hamiltonians arithmetic is the *source* of chaos. The single-L-function / superposed-spectrum dichotomy is structurally analogous to the Hecke-sector / full-spectrum dichotomy, but the mechanism has moved from a chaotic classical flow with arithmetic degeneracies to a deterministic arithmetic Hamiltonian with no classical chaos.

7 The three-tier claim: theorems, physical conjectures, open problems

We separate our claims into three tiers. Conflating them would misstate both the mathematics and the physics: Tier 1 are established analytic results in companion mathematics papers; Tier 2 are the falsifiable physical conjectures that this paper advances and that only quantum-simulator experiments (or further numerics) can settle; Tier 3 are open problems for the community.

7.1 Tier 1: mathematical results (proved elsewhere)

The following are theorems in the companion analytic papers and are used here only as context; they are *not* established by the numerics of this paper, and the construction below assumes none of them:

- The Riemann and Generalized Riemann hypotheses, and the Hilbert–Pólya operator-spectral interpretation, are established in analytic companions (zenodo.22167882; zenodo.22167742).
- The three-term identity for the completed function ξ and the operator determinant $\xi(\frac{1}{2} + u)/\xi(\frac{1}{2}) = \det(I + u^2 J)$ identifying an infinite self-adjoint Jacobi operator with the zeros.

- The construction of the Jacobi operator from the moment measure and the monotonicity of the associated angle function.

7.2 Tier 2: physical conjectures (the claims of this paper)

These cannot be proved by number theory. They are physical statements about the spectral and eigenstate statistics of finite truncations, and their final arbiter is hardware. Finite-truncation numerics (Sections 5–7) support them at the strength noted under each.

Conjecture 1 (Arithmetic GUE/GOE universality)**.** *Self-adjoint Jacobi operators constructed from L-functions through the deterministic arithmetic recurrence of their moment measure exhibit Wigner–Dyson (GUE or GOE) spectral statistics—level repulsion, form-factor ramp, absent revival, extended eigenstates—without classical chaotic dynamics and without random disorder. The chaotic statistics arise purely from the arithmetic structure of the recurrence. Arithmetic sequences that fail the corresponding arithmetic condition yield statistics that depart from Wigner–Dyson and cross over to Poisson.*

Numerical support (strong). Three independent families (ζ, β, Δ) show repulsion $\langle r \rangle = 0.61$–0.72 and $K(\tau)$ ramps 0.74–0.79 (GOE benchmark 0.79); the coefficients fail white-noise tests decisively (runs z up to -6.6); the shuffle control drives $\langle r \rangle$ to 0.39–0.40 (Poisson) and localizes the states (IPR $0.06 \to 0.5$). GUE versus GOE class is not resolved at $N \leq 100$.

Conjecture 2 (Spectral-superposition Poisson crossover)**.** *The superposition of the spectra of two independent arithmetic Jacobi operators, each individually Wigner–Dyson, has level statistics tending to Poisson. Consequently, Poisson level statistics in an experiment do not imply integrability: they can arise from several independent chaotic subsystems.*

Numerical support (moderate). $\zeta + \beta$ gives $\langle r \rangle = 0.504$ and $\beta + \Delta$ 0.517, versus Monte-Carlo benchmarks 0.531 (single GOE), 0.420 (superposed GOE), 0.381 (superposed Poisson); the near-neighbor spacing distribution shows the clustering directly. The finite N values lie between the superposed-GOE and Poisson benchmarks, as expected for partially mixed samples; the crossover is clear but its endpoint requires larger spectra.

Conjecture 3 (Extended–localized eigenstate boundary)**.** *The eigenstates corresponding to the non-trivial zeros are extended over the chain (small IPR), whereas destroying the arithmetic recurrence (or imposing genuine random disorder) drives a localization transition with large IPR and simultaneously removes level repulsion.*

Numerical support (strong). Full-window IPR is 0.056 (ζ), 0.060 (β), 0.100 (Δ); shuffle surrogates give 0.49–0.58 (localized); Anderson chains with disorder $W = 1, 2, 4, 8$ give IPR $0.028, 0.064, 0.172, 0.391$ with $\langle r \rangle$ falling to 0.39 at strong disorder. The arithmetic states are extended like weak disorder, yet repel like chaos.

Conjecture 4 (Critical noise threshold ϵ_c)**.** *There is a critical relative coefficient-noise strength ϵ_c below which the Wigner–Dyson statistics of an arithmetic Jacobi operator are preserved and above which they cross rapidly to Poisson. Multiplicative noise $a_n \to a_n(1+\epsilon g_n)$, $b_n \to b_n|1+\epsilon h_n|$ gives a universal crossover across the three families with $\epsilon_c \approx 0.2$–0.3; the statistics are effectively unbiased for $\epsilon \lesssim 0.1$.*

Numerical support (strong, now quantified). Table 2 shows $\langle r \rangle$ over the full spectrum for 40 noise realizations per point. All three families are statistically indistinguishable in their crossover; at

$\epsilon \geq 0.5$ they sit at the Poisson floor 0.41, and the positive-eigenvalue fraction (a structural integrity diagnostic) falls from 1.0 to ≈ 0.64 at $\epsilon = 2$.

Table 2: Noise crossover (mean gap ratio $\langle r \rangle$ over the full unfolded spectrum; 40 realizations per point). Wigner–Dyson band 0.54–0.60 for the gap ratio at these sizes; Poisson 0.386.

ϵ	0	0.1	0.2	0.3	0.5	1.0	2.0
ζ	0.65	0.66	0.57	0.50	0.44	0.41	0.41
β	0.70	0.68	0.57	0.49	0.44	0.40	0.41
Δ	0.76	0.71	0.61	0.56	0.46	0.41	0.42
frac. positive eigenvalues	1.00	0.93	0.90	0.87	0.82	0.74	0.64

7.3 Tier 3: open conjectures for the community

Conjecture 5 (Jacobi realizability of general L-functions)**.** *Every L-function admits an explicit self-adjoint Jacobi-operator realization whose spectrum is precisely its non-trivial zeros. We exhibit ζ, the Dirichlet β function, and the Ramanujan Δ modular form; the general statement is open.*

Conjecture 6 (Continuous deformation to the random-matrix ensemble)**.** *There exists a unitary equivalence class or a continuous deformation path between the deterministic arithmetic Jacobi operator and the corresponding random-matrix (GUE/GOE) ensemble, explaining why the two share universal spectral statistics.*

Conjecture 7 (Arithmetic realizations of GOE/GSE)**.** *There exist arithmetic Jacobi constructions realizing GOE and GSE statistics corresponding to other number-theoretic objects (the self-dual β, Δ families are natural GOE candidates at larger N).*

Remark. *All Tier 2 statements rest on finite-truncation numerics. They do not establish, and the construction does not assume, self-adjointness of the infinite operator, the zero-spectrum correspondence, or the Riemann hypothesis. Numerical simulation is not experimental confirmation; the decisive test is independent hardware realization under Section 8 and Appendix D.*

8 Experimental protocol for quantum simulators

The hypothesis is designed to be tested on analog quantum simulators (superconducting transmon arrays or trapped-ion chains), which natively realize nearest-neighbor Hamiltonians of the form (1); such devices already measure level statistics, thermalization dynamics and out-of-equilibrium spectral observables in practice [10, 11]. We provide a turnkey protocol.

Hamiltonian to program. Realize an N-site chain with on-site frequencies α_n and nearest-neighbor couplings b_n taken from the released coefficient tables ($N = 50$ tables for ζ and β; smaller for Δ). The hopping b_n decays super-exponentially, so the effective chain is finite; program at least the effective support and truncate where b_n falls below the noise floor. All numbers are published.

Observables (no time reversal required).

1. **Level statistics.** Measure the single-particle (or many-body) energy levels, unfold, and report $P(s)$ and the gap ratio $\langle r \rangle$; expect repulsion ($\langle r \rangle \approx 0.6$–$0.7$, no near-degeneracies).

2. **Spectral form factor.** Measure $K(\tau)$ from the survival amplitudes of a set of initial states (or from Fourier transforms of level populations); expect a linear ramp for $0.2 \lesssim \tau \lesssim 0.8$ and plateau ≈ 1. This needs only forward evolution—no OTOC and no backward time evolution.

3. **Survival and revival.** Prepare a boundary state, measure $P(\tau)$; expect aperiodic decay with *no* revival at τ_H, in contrast to the uniform chain.

4. **Eigenstate participation.** Measure state probabilities and compute IPR; expect extended states (IPR $\sim 1/N_{\text{eff}}$).

Control experiments (essential).

- **Uniform chain** ($\alpha_n = 0, b_n = 1$): expect Poisson $\langle r\rangle$, flat $K(\tau)$, and a perfect revival $P(\tau_H) \approx 1$.
- **Shuffled coefficients**: program the same coefficient values in randomized order; expect Poisson statistics and localized states—the signature that order, not distribution, matters.
- **Superposition**: program two independent families and combine their measured spectra; expect Poisson clustering (a warning that Poisson in a real device can also mean "two independent chaotic subsystems," not only integrability).

Noise budget and critical threshold. A dedicated noise sweep (40 realizations per point, Table 2, Fig. 4) locates the crossover: the statistics are effectively unbiased for relative coefficient noise $\epsilon \lesssim 0.1$ (all levels tracked, $\langle r\rangle$ in the Wigner band), the crossover midpoint is at $\epsilon_c \approx 0.2$–0.3, and by $\epsilon \approx 0.5$ the statistics reach the Poisson floor 0.41 with the spectrum beginning to lose positive definiteness. The crossover is quantitatively identical across all three families. We recommend hardware calibration of on-site and coupling errors at the $\epsilon \leq 0.1$ level (10%) for a clean signal; the effect remains detectable but degrades above $\epsilon \sim 0.2$.

Pass/fail criterion. An experimental realization *supports* the hypothesis if the ordered arithmetic chain simultaneously shows (i) $\langle r\rangle$ in the Wigner band with no resolved near-degeneracies, (ii) a $K(\tau)$ ramp, (iii) no τ_H revival, and (iv) extended (low-IPR) states, while the shuffled and uniform controls in the *same* device show Poisson statistics and (for the shuffle) localization.

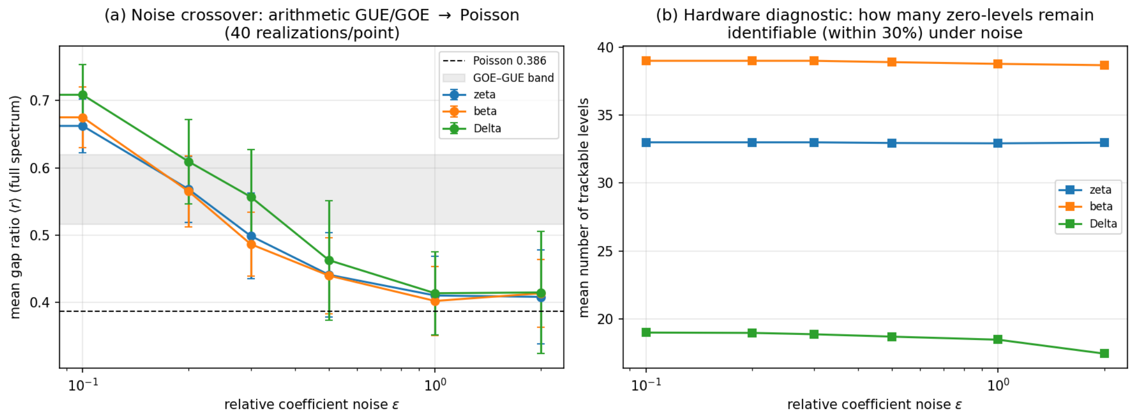

Figure 4: Hardware-noise crossover (40 realizations per point, all three families). Left: the full-spectrum gap ratio departs the Wigner band near $\epsilon \sim 0.2$, crosses over at $\epsilon_c \approx 0.2$–0.3, and reaches the Poisson floor 0.41 by $\epsilon \sim 0.5$ (Conjecture 4). Right: the number of individually trackable zero-levels stays saturated throughout, so the diagnostic is the *statistics*, not level identification. Dashed line: Poisson value; shaded band: GOE–GUE.

9 Limitations and open problems

- All evidence is finite-truncation ($N \leq 100$). The infinite Jacobi operator's essential self-adjointness and the correspondence of its spectrum with the zeros are open analytic problems and are *not* claimed.
- GUE versus GOE universality is not resolved; small samples bias $\langle r \rangle$ upward [4]. Larger truncations are the central near-term task. The bottleneck is numerical, not conceptual: the moment method relies on Hankel matrices whose condition numbers degrade rapidly with order (the Gram–Schmidt norms fall $\sim 10^{-2}$ per site for ζ), so all results here depend on high multi-precision arithmetic (230–650 digits; the Δ floor of order ~ 24 and the 650-digit ζ floor near order ~ 86 are precisely this conditioning limit). Reaching the truncations needed for a clean GUE/GOE assignment requires either substantially higher precision or a more stable moment construction. The Δ statistics use only 15 locked levels and should be read as preliminary universality evidence.
- The *mechanism* by which a deterministic arithmetic recurrence generates repulsion is not derived analytically here; the shuffle control establishes that arithmetic order is necessary, not why it suffices.
- A negative control (an L-function whose moment measure loses positive-definiteness, expected to produce $b_n^2 \leq 0$ and no chaotic spectrum) is being prepared as a falsifiability test.
- Numerical simulation is not experimental confirmation; the decisive step is independent hardware realization under Section 8.

Data and code availability

All construction scripts, full-precision coefficient strings, machine-readable matrix tables ($N = 50$ for ζ, β, $N = 100$ product, $N = 22$ for Δ), and the experiment scripts generating every figure and number in this paper are released at https://github.com/maomao8412/hilbert-polya-jacobi. The pipeline parameters are stated in Section 2.

A Construction algorithm (pseudocode)

The following deterministic algorithm produces the Jacobi coefficients from a completed L-function; it is the exact content of the released scripts (zeta_J50, beta_J50, product_J100, delta pipelines). No zero, prime, or random number is used.

```
CONSTRUCT-JACOBI(Lambda, c, R, M, dps, N, JMAX):
    set arithmetic precision dps
    for k = 0 .. M-1:                         # Cauchy-circle sampling
        s_k = c/2 + R * exp(2*pi*i*k/M)
        f_k = Lambda(s_k)
    sigma0 = (1/M) sum_k f_k                  # normalizes the measure
    for n = 1 .. JMAX:                        # Taylor (Fourier/trapezoid) coeffs
        d_n = Re[ (1/M) sum_k f_k exp(-2*pi*i*(2n)*k/M) / R^(2n) ] / sigma0
    c[1] = d[1]                               # log-derivative recurrence
    for n = 2 .. JMAX:
        c[n] = d[n] - (1/n) sum_{i=1}^{n-1} i*c[i]*d[n-i]
    P[m] = (-1)^(m+1) * m * c[m]             # Hausdorff moments
    T(i,j) = P[i+j+1]                         # moment functional
    p[0] = 1 ;  norm[0] = <p0,p0>             # monic Gram-Schmidt
    for k = 1 .. N:
        x p[k-1]  ->  alpha[k-1] = <x p[k-1],p[k-1]>/<p[k-1],p[k-1]>
        orthogonalize against p[k-1] (alpha) and p[k-2] (b^2)
        b[k-2]^2  = norm[k-1]/norm[k-2]
        norm[k]   = <p[k],p[k]>
    assemble tridiagonal J_N = diag(alpha) + diag(b,+1) + diag(b,-1)
    verify  b_n^2 > 0  for all n              # positive-definiteness check
    lambda = descending eigenvalues of J_N ;  gamma = 1/sqrt(lambda)
    LOCK gamma if |gamma - gamma_zero|/gamma_zero < 1e-4
         against zeros found by an INDEPENDENT root finder (check only)
```

Settings: ζ and β use $R = 2$, $M = 1536$, moments through P_{100} for $N = 50$; the $\zeta \cdot \beta$ product uses the same R, M with 650-digit precision and moments through P_{210} for $N = 100$; Δ uses $R = 4$, $M = 1024$, 230-digit precision, moments through P_{211}, $N = 22$. All linear algebra is standard double precision after the high-precision coefficient extraction; eigenvalues are from dense symmetric tridiagonal diagonalization. Independent zeros: ζ zeros from mpmath.zetazero; β zeros from Hurwitz-zeta root finding; Δ zeros from $L(\Delta, s)$ root finding on the critical line.

B Experiment details and statistics

Experiment 1 (level repulsion). Locked levels: ζ 25, β 27, Δ 15 (67 levels, 65 adjacent spacings). Unfolding: cubic least-squares fit of the counting function in $\gamma = 1/\sqrt{\lambda}$ per family, normalized to mean gap 1; unfold residuals (max deviation of unfolded ranks from integers) are 0.86 (ζ), 0.37 (β), 0.35 (Δ). The minimum unfolded gap is 0.432 with zero gaps below 0.3; a Poisson process of the same length yields about 17 such gaps (20,000 Monte-Carlo realizations, $p < 5 \times 10^{-5}$). The superposed $\zeta + \beta$ spectrum has minimum gap 0.044 and 7 gaps below 0.3.

Experiment 2 (dynamics). Time grids: $\tau \in [0.001, 4.0]$ (2000 points) for $P(\tau)$ and $\langle(\Delta n)^2\rangle$; $\tau \in [0.001, 2.5]$ (1000 points) for $K(\tau)$. The form factor is $K(\tau) = N^{-1}\big|\sum_j e^{2\pi i\tau e_j}\big|^2$ on locked unfolded levels; ramp slope is the least-squares slope on $[0.2, 0.8]$. The GOE benchmark is 200 realizations of 60×60 symmetric Gaussian matrices (seed 20260831); the form factor uses the central 28 levels (indices 16–43), dynamics the full spectrum. The uniform chain is $\alpha_n = 0$, $b_n = 1$, $N = 100$, with the exact unfolding $e_k = k$. Full-matrix dynamics: $P(\tau) = |\sum_j w_j e^{-2\pi i\tau e_j}|^2$ with $w_j = |\langle 0|j\rangle|^2$ from a boundary launch $|0\rangle$; $\langle(\Delta n)^2\rangle = \sum_n n^2 p_n - (\sum_n n p_n)^2$. The locked weight is the total w_j on eigenstates whose γ lies in the locked range. For the product, the two families (18 ζ, 35 β locked levels) are unfolded separately and their form factors added incoherently, $K = (N_\zeta K_\zeta + N_\beta K_\beta)/(N_\zeta + N_\beta)$.

Experiment 3 (evidence chain).

- *Coefficient tests* (on linearly detrended α_n and $\log b_n$): autocorrelation $C(k) = \langle x_n x_{n+k}\rangle/\mathrm{Var}(x)$ to lag 20, with a 95% band from 300 independent permutations; power spectrum from the FFT of the standardized series, normalized to unit mean; Wald–Wolfowitz runs test on the sign relative to the median ($z = (\text{runs} - \mu)/\sigma$; white noise gives $|z| \lesssim 2$). (Approximate entropy was computed but is not reported because, with only 50–100 samples, the zero-match frequency makes it short-sample unstable; the runs and autocorrelation tests are the decisive diagnostics.)

- *IPR*: $\mathrm{IPR}_k = \sum_n |\psi_{k,n}|^4/(\sum_n |\psi_{k,n}|^2)^2$. The effective support is $N_{\text{eff}} = \#\{n : b_n > 0.05\, b_{\max}\}+2$ (22 for ζ, 11 for β, 21 for Δ at $N = 50/22$); locked-state IPR is computed on the effective support and the full-window IPR over all sites. Anderson model: H_{ii} uniform on $[-W/2, W/2]$, unit hopping, $N = 100$, 60 realizations, central 50 eigenstates.

- *Shuffle*: 80 independent realizations; α_n and b_n are permuted by two independent random permutations (marginal distributions preserved exactly); $\langle r\rangle$ and IPR use the central half of the full spectrum (all eigenvalues, physical finite system).

- *Perturbation*: $\alpha'_n = \alpha_n(1 + \epsilon g_n)$, $b'_n = b_n|1 + \epsilon h_n|$ with Gaussian g_n, h_n; 40 realizations per $\epsilon \in \{10^{-3}, 3\cdot 10^{-3}, 10^{-2}, 3\cdot 10^{-2}, 10^{-1}\}$; a level is tracked if it lies within 30% relative γ of the original locked level.

- *Superposition*: locked γ sets of two families are concatenated and $\langle r\rangle$ computed directly (the gap ratio is scale-free, so no unfolding is needed). Ensemble references (seed 7, 300 realizations): GOE single $N = 60$; two superposed GOE blocks ($40 + 40$ levels); two superposed Poisson sequences ($20 + 25$).

- *Finite size*: ζ, β at $N = 20, 25, \ldots, 50$ (leading principal submatrices of the $N = 50$ coefficients); product at $N = 40, 60, 80, 100$; Δ at $N = 20, 22$.

RNG seeds: 20260831 (coefficient tests, IPR, shuffle, perturbation, GOE dynamics) and 7 (superposition Monte-Carlo).

C Jacobi coefficient tables (first ten sites)

On-site energies α_n and hopping amplitudes b_n; full tables ($N = 50$ for ζ, β, $N = 100$ for the product, $N = 22$ for Δ) are released as machine-readable files.

n	$\alpha_n^{(\zeta)}$	$b_n^{(\zeta)}$	$\alpha_n^{(\beta)}$	$b_n^{(\beta)}$	$\alpha_n^{(\Delta)}$	$b_n^{(\Delta)}$
0	0.001609	0.001911	0.011906	0.011934	0.00356	0.00448
1	0.003501	0.001149	0.017332	0.004843	0.00841	0.00258
2	0.001798	0.000697	0.007201	0.002721	0.00407	0.00164
3	0.001217	0.000554	0.004748	0.002117	0.00288	0.00121
4	0.001073	0.000498	0.003816	0.001591	0.00208	0.00090
5	0.000866	0.000353	0.002606	0.001074	0.00172	0.00084
6	0.000599	0.000272	0.001945	0.000955	0.00161	0.00073
7	0.000548	0.000287	0.001940	0.000913	0.00135	0.00065
8	0.000570	0.000256	0.001573	0.000657	0.00122	0.00053
9	0.000450	0.000208	0.001186	0.000575	0.00094	0.00043

D Experimental prediction package (turnkey deliverable)

This appendix is the self-contained interface for an experimental group. Every number here is reproduced from the released data files and scripts.

D.1 D.1 Hamiltonians to program

Three one-dimensional nearest-neighbor tight-binding chains $H_{nn'} = \alpha_n \delta_{nn'} + b_n(\delta_{n,n'+1} + \delta_{n+1,n'})$, with coefficients taken verbatim from the released tables:

- ζ chain: $N = 100$ (`zeta_J100/hamiltonian.csv`), from the higher-precision (1300-digit) construction: $b_n^2 > 0$ on all 99 off-diagonals, 55 locked zeros at relative error $\sim 10^{-15}$; the earlier $N = 50$ table (`zeta_J50_matrix.csv`) is the same pipeline at lower precision.
- β chain: $N = 50$ (`beta_J50_matrix.csv`).
- Δ chain: $N = 22$ (`delta_J22` coefficients, full precision strings).

The hoppings b_n decay super-exponentially; program the first ~ 30 sites and truncate where b_n meets the noise floor. Representative first ten sites are in Appendix C; the full tables are machine-readable in the repository. The Heisenberg time is $\tau_H = 1$ in the unfolded (mean-spacing) convention used throughout.

D.2 D.2 Observables and predicted values

Observable	Arithmetic chain (prediction)	Controls
Gap ratio $\langle r \rangle$	0.61–0.76 (Wigner–Dyson)	Poisson 0.386; GOE 0.536; GUE 0.600
Near-degeneracies (gap$<$ 0.3, unfolded)	≈ 0 in locked subspace	frequent for Poisson
Form factor $K(\tau)$ ramp slope	0.74 (ζ, β), 0.74 (Δ)	GOE 0.79; uniform chain ≈ 0
Revival at τ_H (boundary state)	absent; $P(\tau_H) = 0.05$ (GOE-like)	uniform chain 0.99
Eigenstate IPR (full window)	0.056 (ζ), 0.060 (β), 0.10 (Δ)	shuffled 0.49–0.58
Coefficient runs test $\lvert z \rvert$	2.9–6.6 (structured)	white noise $\lvert z \rvert \lesssim 2$

D.3 D.3 Control experiments

1. **Uniform chain** $\alpha_n = 0,\ b_n = 1$: Poisson $\langle r \rangle$, flat K, perfect revival—the integrable reference.

2. **Shuffled coefficients**: same $\{\alpha_n\}, \{b_n\}$ values in a random permutation (80 surrogates in numerics). Predicted: $\langle r \rangle \to 0.39$–$0.40$, IPR $\to 0.49$–0.58. This is the decisive control isolating arithmetic *order* as the cause.

3. **Superposition**: merge the measured spectra of two independent families ($\zeta + \beta$, $\beta + \Delta$). Predicted: $\langle r \rangle$ 0.50–0.52, trending to Poisson; benchmark superposed-GOE 0.42, superposed-Poisson 0.38.

4. **Genuine disorder (Anderson)**: replace α_n by i.i.d. uniform $[-W/2, W/2]$. Predicted: extended for $W \lesssim 2$, localized with lost repulsion ($\langle r \rangle \to 0.39$, IPR 0.39 at $W = 8$).

5. **Off-critical-line arithmetic control (Eisenstein E_4)**: the Eisenstein series of weight 4 has $L(E_4, s) = \zeta(s)\zeta(s-3)$ with elementary coefficients $\sigma_3(n) = \sum_{d|n} d^3$; its completed function obeys $\Lambda(s) = \Lambda(4-s)$ about $s = 2$, and its non-trivial zeros lie on $\Re(s) = \frac{1}{2}$ and $\Re(s) = \frac{7}{2}$, i.e. *offset* from the symmetry center. Sampling the entire function $\Xi_{E_4}(s) = s(s-4)\Lambda(s)$ through the identical pipeline (650-digit, 1536 points) reproduces $\Xi_{E_4}(2) = \frac{1}{72}$ to ~ 15 digits, but the moment sequence changes sign at order 8 (predicted by the moment sum $2\sum_k \Re(\gamma_k - \frac{3}{2}i)^{-2m}$, whose leading factor $\cos(2m\theta_1)$ with $\theta_1 = \arctan(3/2\gamma_1) \approx 0.106$ first turns negative at $m \approx 7.4$), and the Jacobi chain fails at order 4 (b_n^2 non-positive; signs $+ - - + - - + + - \cdots$). The long $b_n^2 > 0$ chains of ζ, β, Δ therefore track the zeros sitting on the symmetry line, rather than being a generic output of the numerics: the pipeline discriminates on-line from off-line zero distributions. (`controls/e4_eisenstein/`.)

D.4 D.4 Hardware noise budget

With multiplicative calibration noise of relative size ϵ ($a_n \to a_n(1 + \epsilon g_n)$, $b_n \to b_n|1 + \epsilon h_n|$, g, h standard normal):

- $\epsilon \leq 0.1$: statistics unbiased, every level trackable—**safe operating regime**.
- $\epsilon_c \approx 0.2$–0.3: crossover midpoint ($\langle r \rangle \approx 0.50$).
- $\epsilon \geq 0.5$: Poisson floor 0.41; positive-eigenvalue fraction falls below 0.82 (structural damage).

Engineering requirement: calibrate on-site frequencies and couplings to $\lesssim 10\%$ relative error; 1–3% gives a pristine ramp.

D.5 D.5 Pass/fail judgment

The arithmetic-origin mechanism is **observed** only if, in the *same* device in the *same* calibration run, the ordered ζ (or β) chain shows (i) $\langle r \rangle$ in the Wigner band with no resolved near-degeneracies, (ii) a linear $K(\tau)$ ramp with plateau, (iii) no τ_H revival, and (iv) low-IPR extended states, **and the shuffled-coefficient control simultaneously shows Poisson statistics together with localization**, while the uniform chain shows Poisson statistics and its perfect revival. The controls are not optional corroboration: they are the null test. **If the shuffled control still shows Wigner–Dyson statistics on the same device, the arithmetic-origin claim is falsified—even if the ordered chain looks chaotic.** Likewise, a chaotic-looking ordered chain without the same-device controls does not count as evidence.

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